

HAMZA ZEESHAN AHMED

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EDUCATION

Information Technology University (ITU), Lahore, Pakistan

Expected Aug 2028

B.S. in Artificial Intelligence

- CGPA: 3.49 / 4.00
- Relevant Coursework: Data Structures & Algorithms in C++, OOP, CNN & Logistic Regression (Programming for AI), Database Management Systems, Artificial Intelligence, Design & Analysis of Algorithms, Probability & Statistics, Linear Algebra

Government College University (GCU), Lahore, Pakistan

2024

FSc Pre-Engineering

PROJECTS

AI Handwriting Generation System

[GitHub Repo](#)

Machine Learning Developer

2026

- Designed a 3-layer CNN that decomposed each character into 6 Bézier curves (48 control points), training on team collected samples to synthesize personalised, font-free handwriting with natural stroke variation.
- Used iterative model refinement across 5 approaches to achieve MSE 0.0042 (target < 0.005) and SSIM 0.851 over 105 epochs, with each CNN layer targeting a distinct abstraction: edge detection → curvature → stylistic traits.

Cinelog — Full-Stack Film Diary Platform

[GitHub Repo](#)

Full-Stack Developer

2025

- Used C++ to build all backend data storage, B-Trees, Tries, and Hashmaps, entirely from scratch with zero STL containers, enabling user profiles, film logs, reviews, and watchlists through self built data structures.
- Used free-tier EC2 AWS to deploy the platform end-to-end, demonstrating full ownership from custom low-level data structures to cloud infrastructure, mirroring the architecture of a production film-diary application.

Chess Engine — OOP-Based Graphical Chess

[Play Here](#)

Game Developer

2025

- Used C++ and Raylib to implement a complete chess engine with full legal-move validation, castling, promotion, and checkmate detection, with a chess.com-inspired UI deployed and playable in-browser via GitHub Pages.
- Used OOP principles, separate classes for board state and each piece type, to build a clean, extensible architecture that isolates game logic from rendering, simplifying future feature additions.

Linear & Logistic Regression from Scratch

Machine Learning Developer

2026

- Used NumPy and Pandas exclusively, no scikit-learn, to implement gradient descent, cost functions, and evaluation metrics for both regression variants, demonstrating first-principles understanding of ML optimization.

SKILLS

Programming: C++, Python, SQL, Dart, 8086 Assembly

Tools & Frameworks: Git/GitHub, PostgreSQL, MySQL, MongoDB, Redis, AWS, Flutter, Vercel, Linux, NASM

AI / ML: PyTorch, NumPy, Pandas, Hugging Face

Concepts: Data Structures & Algorithms, OOP, Full-Stack Development, CNN, Regression Modelling, DBMS

ACTIVITIES AND LEADERSHIP

ITU Sports Society

Aug 2024 – May 2025

Head Video Editor

- Managed all video production and publishing for ITU Sports Society events across the academic year.

Coderush, ITU Competitive Programming Contest

2025 & 2026

Contestant

- Participated in competitive programming competitions organized by GDG ITU on campus.